

Ph.D. Qualifying Exam 2023: Statistical Inference

July 3, 2024
10 AM to 2 PM

Rules: This is a four hour exam consisting of TWO PARTS. PART I is based on Stat 593 and contains **four independent problems**. Part II is based on Stat 594 and contains **one problem with six questions**. PART I is closed notes, closed book. For PART II, you may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “information sheet” for Stat 594. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly in the space provided after each problem, preferably using pens, leaving an impression deep enough for the papers to be scanned. Good luck!

PART I

Please read the following general instructions carefully.

1. This is a closed-book and closed-note exam.
2. The exam is part of PhD qualifying exam on statistical inference.
3. Please work on your solutions **independently**.
4. Please show the details of your derivations.
5. Partial credit will be given if you make some reasonable progress but cannot solve the problem completely.
6. Please only include the derivations that you believe are correct. Any wrong argument will result in a grade deduction.
7. ♣ indicates that the problem might be slightly more challenging than others.

Problem 1. MLE and Hypothesis Testing for Exponential Family.

Consider that we have access to the i.i.d. data $X_1, \dots, X_n$ following the one-parameter exponential family,

$$p(x|\theta) = h(x) \exp\{\eta(\theta)T(x) - B(\theta)\}, \quad \text{for } x \in \mathcal{X}$$

with $\mathcal{X}$ denoting the support. The exponential family has the natural form as

$$p(x|\eta) = h(x) \exp\{\eta T(x) - A(\eta)\},$$

where η is the canonical parameter.

- (a) Identify the sufficient statistics for θ and compute the MLE estimator for θ .

(b) State the expression for $A(\eta)$ in terms of θ .

- (c) Assume that $T(x)$ is not a constant function over the support $\mathcal{X}$ and $\eta(\theta)$ is a one-to-one mapping. Show that the MLE is unique. ♣
(Hint: show that $A(\eta)$ is strictly convex.)

For the remaining problems, assume that $\eta(\theta)$ is strictly monotone increasing in θ and $\sum_{i=1}^n T(X_i)$ is a continuous random variable.

- (d) Construct the most powerful test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ with $\theta_1 > \theta_0$.

- (e) Construct the uniformly most powerful test for $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$. (You need to justify why such a test is uniformly most powerful.)

- (f) Prove or disprove the existence of uniformly most powerful test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Problem 2. MLE and One-step Estimator. Consider that $X_1, \dots, X_n$ are i.i.d. with the density function $p_\theta(\cdot)$. Assume the required regularity conditions are satisfied.

1. Define the MLE and prove the asymptotic limiting distribution of MLE.
(You can assume the consistency of MLE.)

2. State the one-step estimator and prove it has the same asymptotic distribution as the MLE estimator.

Problem 3. Statistical Inference in High-dimensional Regression.

Consider the high-dimensional sparse linear regression

$$y_i = X_i^\top \beta + \varepsilon_i \quad \text{for } 1 \leq i \leq n$$

where $\beta \in \mathbb{R}^p$, $\|\beta\|_0 \leq k$ and ε_i follows i.i.d normal random variable with mean zero and variance σ^2 .

1. Define the Lasso estimator $\hat{\beta}$ and discuss the challenge of statistical inference with the Lasso estimator.

2. Write down one form of the debiased Lasso estimator for the j -th regression coefficient β_j with $1 \leq j \leq p$ and establish the limiting distribution of your chosen debiased Lasso estimator.

3. Construct a confidence interval for β_j , where you need to estimate σ^2 in a data-dependent way.

Problem 4. Regression with hidden confounders. Consider the following structural equation model

$$Y_i \leftarrow D_i\beta + H_i\psi + e_i,$$

where Y_i and D_i denote the outcome and treatment variables, H_i denotes the hidden variable, and e_i denotes the independent random error. For simplicity, we assume D_i , H_i , and e_i are centered random variables.

1. Explain when the OLS estimator is a consistent estimator of β .

2. Define the instrumental variable and propose a consistent estimator of β using an instrumental variable.

PART II

Suppose that $X_1, \dots, X_n$ are independent and identically distributed from a p -variate normal distribution with mean vector 0 and unknown variance matrix Σ . Assume that the smallest eigenvalue of Σ is lower bounded by a constant $\sigma_0 > 0$ (hence Σ is non-singular), and the largest eigenvalue of Σ is upper bounded by a constant $\sigma_1 > 0$. For a fixed vector $\xi^* \in \mathbb{R}^p$, it is of interest to estimate $u^* = \Gamma \xi^*$, where $\Gamma = \Sigma^{-1}$ is also called the precision matrix.

For p much less than n , a simple estimator of u^* is $\hat{u} = \hat{\Sigma}^{-1} \xi^*$, where $\hat{\Sigma} = \mathbf{X}^T \mathbf{X} / n$, and $\mathbf{X}$ is the $n \times p$ matrix, $\mathbf{X} = (X_1, \dots, X_n)^T$. Equivalently, $\hat{u}$ can be determined as

$$\hat{u} = \operatorname{argmin}_{u \in \mathbb{R}^p} \left\{ \frac{1}{2} u^T \hat{\Sigma} u - u^T \xi^* \right\}.$$

For p close to or larger than n , we consider a Lasso penalized estimator, defined as

$$\hat{u}_\lambda = \operatorname{argmin}_{u \in \mathbb{R}^p} \left\{ \frac{1}{2} u^T \hat{\Sigma} u - u^T \xi^* + \lambda \|u\|_1 \right\}, \quad (1)$$

where $\|\cdot\|_1$ denotes the L_1 norm and $\lambda > 0$ is a tuning parameter. The Lasso penalized estimator is expected to perform well in the setting where the true value u^* is sparse even with p larger than n . In this problem, you are asked to study convergence of $\hat{u}_\lambda$ to u^* .

For each question, you may use the results from the preceding questions, whether or not you have answered those questions. You may also use the following result.

Suppose that the pairs (Z_{i1}, Z_{i2}) , $i = 1, \dots, n$, are independent, and Z_{ij} is mean-zero and sub-Gaussian with scale parameter τ_j for $i = 1, \dots, n$, $j = 1, 2$. Then, with $Y_i = Z_{i1} Z_{i2}$,

$$P \left\{ \frac{1}{n} \left| \sum_{i=1}^n (Y_i - E(Y_i)) \right| \geq y \right\} \leq 2 \exp \left\{ -C_0^{-1} n \min \left(\frac{y^2}{(\tau_1 \tau_2)^2}, \frac{y}{\tau_1 \tau_2} \right) \right\}, \quad y \geq 0, \quad (2)$$

where $C_0 > 0$ is an absolute constant. Recall that a random variable Z is sub-Gaussian with scale parameter τ if

$$E(e^{\lambda(Z - E(Z))}) \leq e^{\frac{\tau^2 \lambda^2}{2}}, \quad \lambda \in \mathbb{R}.$$

1) Show that $u^{*\top}X_i$ is sub-Gaussian with scale parameter $\sqrt{\sigma_1}\|u^*\|_2$, where $\|\cdot\|_2$ denotes the L_2 norm.

2) Let $\Omega_0 = \Omega_0(\lambda_0)$ be the event that

$$\|\hat{\Sigma}u^* - \xi^*\|_\infty \leq \lambda_0,$$

where $\|\cdot\|_\infty$ denotes the L_∞ norm. For any $\varepsilon \in (0, 1)$, find λ_0 , depending on (n, p, ε) , such that $P(\Omega_0(\lambda_0)) \geq 1 - 2\varepsilon$. The dependency of λ_0 on (n, p, ε) will be suppressed in the notation. [Hint: Use result (2).]

3) Show that for any $u \in \mathbb{R}^p$,

$$\frac{1}{2n} \|\mathbf{X}(\hat{u}_\lambda - u)\|_2^2 + (\hat{\Sigma}u - \xi^*)^\top (\hat{u}_\lambda - u) + \lambda \|\hat{u}_\lambda\|_1 \leq \lambda \|u\|_1.$$

4) Take $\lambda = A_0\lambda_0$ in (1) for a constant $A_0 > 1$. Show that in the event $\Omega_0(\lambda_0)$.

$$\frac{1}{2n} \|\mathbf{X}(\hat{u}_\lambda - u^*)\|_2^2 + (A_0 - 1)\lambda_0 \|\hat{u}_\lambda - u^*\|_1 \leq 2A_0\lambda_0 \sum_{j \in S^*} |\hat{u}_{\lambda,j} - u_j^*|,$$

where u_j denotes the j th component of $u \in \mathbb{R}^p$ (hence $\hat{u}_{\lambda,j}$ is the j th component of $\hat{u}_\lambda$), and $S^* = \{1 \leq j \leq p : u_j^* \neq 0\}$.

5) Let Ω_{Comp} be the event that the following empirical compatibility condition holds for $\hat{\Sigma}$: for some constants $\nu_0 > 0$ and $\eta_0 > 1$, we have

$$\nu_0^2 \left(\sum_{j \in S^*} |b_j| \right)^2 \leq |S^*| (b^\top \hat{\Sigma} b),$$

for any vector $b = (b_1, \dots, b_p)^\top$ satisfying the cone condition

$$\sum_{j \notin S^*} |b_j| \leq \eta_0 \sum_{j \in S^*} |b_j|,$$

where $|S^*|$ denotes the cardinality of S^* . Show that in the event $\Omega_0(\lambda_0) \cap \Omega_{\text{Comp}}$, for a suitable choice of A_0 ,

$$\begin{aligned} \frac{1}{n} \|\mathbf{X}(\hat{u}_\lambda - u^*)\|_2^2 &\leq M_0 |S^*| \lambda_0^2, \\ \|\hat{u}_\lambda - u^*\|_1 &\leq M_0 |S^*| \lambda_0, \end{aligned} \tag{3}$$

where $M_0 > 0$ is a constant, depending on (A_0, ν_0, η_0) but not on (n, p, ε) .

6) Outline how you may show that under certain conditions, $P(\Omega_{\text{Comp}}) \geq 1 - 2\varepsilon$. You do not need to provide all details, but please indicate your main steps and the associated conditions.

To conclude, combining the results from **2), 5), 6)** shows that the error bounds in **(3)** hold with probability at least $1 - 4\varepsilon$.